

Quarterly Business Review (QBR) Summary

Executive Summary

As of 2025-10-05, the SaaS web application runs on EKS in ap-south-1 with ALB ingress, HPA (2–10 replicas), and Cluster Autoscaler. The stack targets scalability, performance, reliability, and cost efficiency.

Current Architecture

- EKS 1.29, single cluster, private+public subnets
- Managed Node Group (t3.small, SPOT), min=1, desired=2, max=5
- ALB (internet-facing) via AWS Load Balancer Controller
- Stateless Python service, 2 replicas, readiness/liveness probes

Performance & Reliability

HPA scales pods at 50% CPU; ALB health checks ensure seamless rollouts. Readiness probes gate traffic to ready pods. Cluster Autoscaler scales nodes to match workload.

Cost Posture

SPOT capacity yields significant savings with a small baseline. Single NAT Gateway used initially. Logging retained minimally. Requests/limits tuned for efficient bin-packing.

Recommendations (Next Quarter)

- 1) Enable HTTPS + AWS WAF managed rules
- 2) Add CI (GitHub Actions) with security scans
- 3) Introduce Prometheus/Grafana or CloudWatch Container Insights
- 4) Consider Graviton (t4g.*) nodes with multi-arch images
- 5) Implement PDBs and custom metrics via KEDA

Risks & Mitigations

- SPOT interruptions → keep 1 on-demand node and PDBs
- Single NAT GW → add multi-AZ NAT for higher availability later
- Secrets → move to AWS Secrets Manager + CSI driver
- Observability gaps → dashboards, SLOs, and alerting